$V \otimes W$
$V \oplus W$

$E\ f(g)$ expected value $1/|G| \sum_{g \in G} f(g)$
$g \in G$

==========

=== Representation theory of finite groups ==

https://en.wikipedia.org/wiki/Representation_theory_of_finite_groups

Linear representation (ρ, V) of G
Subrepresentation
Irreducible representation
Irreducible representation: https://en.wikipedia.org/wiki/Irreducible_representation
Let V be a K-vector space.
Let ρ be a map

$\rho: G \to GL(V)$

$\rho(g): V \to V$
$\rho(gg') = \rho(g)\ \rho(g')$ for all $g,g' \in G$

(ρ, V) is a **linear representation** of G.
V is a **representation space** of G
$\dim(\rho)$ is the **dimension** of the representation.

A linear representation is simply a case of a group action.
i.e. a **group action** on **vector space V by linear transformation ρ(s)**.

$*: G \times V \to V$ as group action
$g * v = \rho(g)\ v$

$g * = \rho(g): V \to V \in GL(V)$ representation

Two representations (ρ, V) and (ρ', V') may be related by a **linear map of vector spaces V → V'**.
i.e. let $T: V \to V'$ be a linear map such that

$\mathbf{T \circ \rho = \rho' \circ T}$
$T(\rho g\ v) = \rho' g\ (Tv)$ for any $v \in V$

equivalently

$\mathbf{T = \rho' \circ T \circ \rho^{-1}}$

T is called an **equivariant map** or **G-linear map** of ρ and ρ'.

Identity representation 1 aka identity character.
Regular representation
https://en.wikipedia.org/wiki/Regular_representation
Two canonical representations that always exist are the **identity character 1** and **regular representation R**.
The identity character is the simplest 1-dimensional representation

$1: G \to GL(C)$
$1(g) = 1 \quad$ for all $g \in G$

$1(g): C \to C$
$1(g)\, x = x$

Clearly, 1 is **irreducible**.

A **free representation** of the group itself
Let F[G] be the free vector space generated by G
i.e. elements are **free sum i.e. formal sum** of $g \in G$

$$\sum_{x \in G} kx\, x \qquad \text{for } kx \in F$$

The (left) regular representation just **shifts** the sum by $g \in G$

$R: G \to GL(F[G])$
$R(g): F[G] \to F[G]$
$R(g) = g$

Applied on $x \in F[G]$

$R(g): F[G] \to F[G]$
$R(g)(x) = gx$

Applied on any element

$R(g)(\sum kx\, x) = \sum kx\, gx$

Average values
A recurring theme is to distribute an effect over all G by averaging

Average E [ρ(g)]
The average

$P: V \to V$
$$P = \mathop{E}_{g \in G} [\rho(g)]$$

is a **projection** into the **fixed points of** ρ.
i.e.

projection $\quad P^2 = P$
fixed points $\quad$ **Im P = Fix(ρ)**
$= \{ v \in V: \rho(g)v = v \}$

For any $h \in G$

$\rho(h)\, Pv = \rho(h)\, \mathop{E}_{g} [\rho(g)v]$
$= \mathop{E}_{g} [\rho(hg)v]$

$= Pv$

$\Rightarrow \text{Im } P \subseteq \text{Fix}(\rho)$

and conversely, for any $v \in \text{Fix}(\rho)$

$Pv = Eg\ [\rho(g)v]$

$= Eg\ [v]$

$= v$

$\Rightarrow \text{Fix}(\rho) \subseteq \text{Im } P$

Hence Imp P = Fix(ρ). Also

$P^2v = Pv$ since $Pv \in \text{Fix}(\rho)$

G-linear E[ρ'(g) ∘ T ∘ ρ(g^-1)]

Let T: V → V' be any linear map and let

T'g: V → V'
T'g = ρ'(g) ∘ T ∘ ρ(g^-1)

T': V → V'
T' = $\mathbf{\underset{g \in G}{E}[T'g]}$

$\mathbf{= \underset{g \in G}{E}[\rho'(g) \circ T \circ \rho(g^{-1})]}$

Then T' is always a **G-linear map:**

$\mathbf{T' \circ \rho(h)} = \underset{g \in G}{E}[\rho'(g) \circ T \circ \rho(g^{-1} h)]$

$= \underset{g' \in G}{E}[\rho'(hg') \circ T \circ \rho(g'^{-1})]$ let g' = h^-1g

$= \mathbf{\rho'(h) \circ T'}$

When ρ' = ρ, then T' can be understood as a **conjugation action**:

T^ρ(g): V → V
T^ρ(g) = ρ(g) T ρ(g^-1)

Common representations

Some common representations

character

A **character** i.e. group homomorphism f: G → C
is a **1-dimensional** representation on C

ρ: G → GL(C)
ρ(g)x = f(g)x for any x ∈ C

with **character** equal to itself

$\chi\rho$ = f since ρ(g) has eigenvalue f(g)
=> tr(ρ(g)) = f(g)

dual representation (ρ*, V*) of (ρ, V)

Just **composition** of f ∈ V* with ρ(g)

ρ*: G → GL(V*)

ρ*(g): V* → V*
ρ*(g)f = f ∘ ρ(g^-1) for any f: V → F ∈ V*

Applying on v ∈ V

ρ*(g)f: V → F
ρ*(g)f (v) = f (ρ(g^-1)v)

This preserves the **duality**

ρ*(g)f ∘ ρ(g) = f ∘ ρ(g^-1) ∘ ρ(g)
= f

direct sum
tensor product

Let ρ1: G1 → GL(V1), ρ2: G2 → GL(V2) be representations.
direct sum
ρ1⊕ρ2: G1⊕G2 → GL(V1⊕V2)
ρ1⊕ρ2(g1⊕g2) = ρ1(g1) ⊕ ρ2(g2)

tensor product
ρ1⊗ρ2: G1⊗G2 → GL(V1⊗V2)
ρ1⊗ρ2(g1⊗g2) = ρ1(g1) ⊗ ρ2(g2)

note: for any f ∈ GL(V1), f ∈ GL(V2)

f⊕g(v⊕w) = f(v)⊕g(w) ∈ V⊕W
f⊗g(v⊗w) = f(v)⊗g(w) ∈ V⊗W

<u>Irreducible representations</u>
Let **subspace W ⊆ V**
ρ is **G-invariant on W** if

ρ(g) w ∈ W for all w ∈ W

Hence the **subrepresentation** (ρ|W, W) is well-defined.

ρ|W: W → W

Every representation ρ has two **trivial subrepresentations**:

(ρ, V) itself

(ρ0, {0}) **trivial** vector space
i.e. just the 0 vector
ρ(g) 0 = 0

A representation is **irreducible** if it has no proper subrepresentations
i.e. no non-trivial representations.

The simplest irreducible representation is the identity character **1: G → GL(C)**

Maschke's Theorem: decomposition
Let (ρ, V) be a representation of G.
If ρ has a subrepresentation in a subspace W of V, then ρ has a subrepresentation in some complement W'.
More precisely

V = W ⊕ W'
ρ = ρ|W ⊕ ρ|W'

Hence any representation ρ can be decomposed into a sum of irreducible representations

V = W1 ⊕ ... ⊕ Wn
ρ **=** ρ**|W1 ⊕ ... ⊕** ρ**|Wn** ρ|Wi irreducible

==========
Proof:
Let p: V → V be a projection into W.
i.e. p^2 = p and pw = w

The idea is just to create a projection p' from p.
Let

p': V → V
p' = E[p'^ρ(g)]

Step 1) p' is a **projection into W**:
a) p' preserves W i.e. p'(w) = w for any w ∈ W

p^ρ(g) w = ρ(g) p ρ(g)^-1 w
= w since ρ(g)^-1 w ∈ W and pw' = w'

=> p'(w) = E [p^ρ(g)w]
= E [w]
= w

b) p'^2 = p'
For any v ∈ V

p^ρ(g) v = ρ(g) p ρ(g)^-1 v ∈ W
since p project into W and preserves w

=> p'(v) = 1/|G| ∑ p^ρ(g) v ∈ W

=> p'(p'(v)) = p'(v) since p' preserves W

Thus W' = ker p' is a complement of W.
Step 2) W' is G-invariant
Let v ∈ W'. For any h ∈ G

p'^ρ(h^-1) (v) = p'(v) = 0

=> p'(ρ(h)) v = 0

*) **Decomposition**
Simply by induction on the **dimension** of V.

case: dim V = 1
Trivially irreducible.

case: dim V = n
If ρ is irreducible, then there is nothing to do.
If ρ is invariant on some subspace W ⊆ V, then by the theorem,

V = W ⊕ W'

=> dim W, dim W' < dim V
    so
      W = W1 ⊕ ... ⊕ Wn ρ|W' irreducible
      W' = W1' ⊕ ... ⊕ Wm' ρ|Wi' irredudible

=> V = (W1 ⊕ ... ⊕ Wn) ⊕ (W1' ⊕ ... ⊕ Wm')

QED
==========

Schur's Lemma: uniqueness
https://en.wikipedia.org/wiki/Schur's_lemma
Let (ρ, V) and (ρ', V') be irreducible representation of G.
Then any **non-trivial** map between ρ and ρ' between must be an isomorphism

i.e. for any T: V → V'

T = ρ'^-1 ∘ **T** ∘ ρ => T is an **isomorphism**
T ≠ 0

Furthermore,

T = T^ρ <=> T is a **scalar** multiple
i.e. **T =** λ**I**

More generally, for any linear map T: V → V', let

T': V → V'
T' = E[ρ**'^-1(g) ∘ T ∘** ρ(g)**]**
  g∈G

Then T' is G-linear and hence

case: ρ'**,** ρ not isomorphic **T' = 0**

case: ρ' **=** ρ **T' = tr(T) / |G| I**

since E[T^ρ(g)] = λI by above, so

taking the trace

=> |G|λ = tr(λI) = E[tr(T^ρ(g))] =

E[tr(T)] = tr(T)

=> λ = tr(T) / |G|

In matrix terms:
Let

{aij} i,j = 1...n of ρ
{bst} s,t =1...m of ρ'

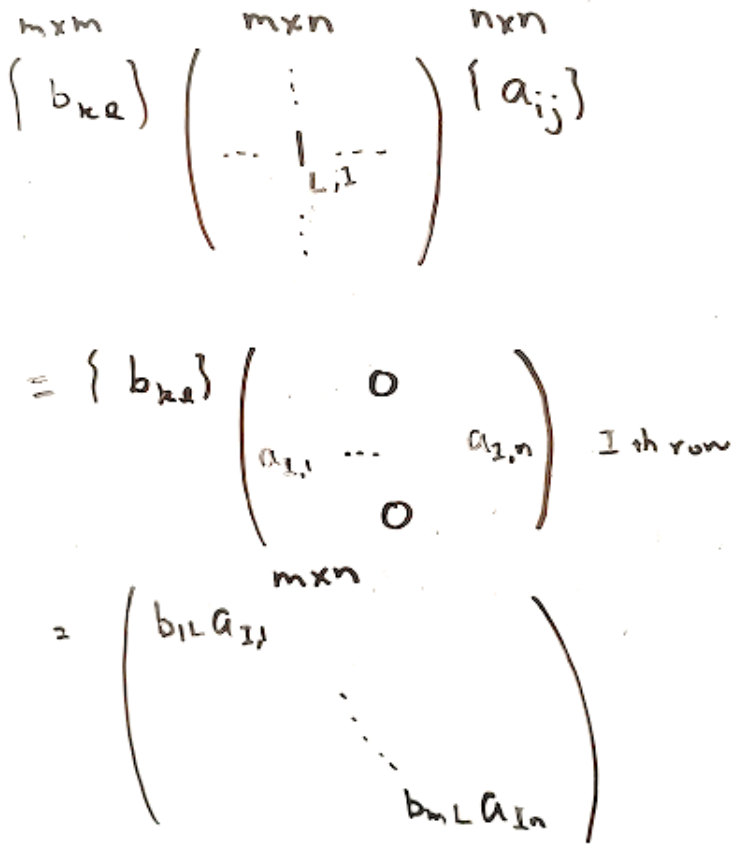
For any i0, t0

1t0,i0 = mxn matrix with 1 in position (t0,i0), 0 otherwise

Then

T' = E[ρ'(g^-1) ∘ 1t0,i0 ∘ ρ(g)]

= { **1/|G|** ∑ **bs,t0 (g^-1) ai0,j(g)** } (k,j) th term
for s = 1...m, j = 1...n



Hence, for any i,j, s,t

case: ρ'**,** ρ not isomorphic $b_{s,t}^{-1} \bullet a_{l,j} = \sum_{g \in G} b_{s,t}(g^{-1})\, a_{i,j}(g) = 0$

case: ρ' **=** ρ $a_{s,t}^{-1} \bullet a_{l,j} = \sum_{g \in G} a_{s,t}(g^{-1})\, a_{l,j}(g)$

$$= \begin{cases} |G|/n & \text{if } j=k \\ 0 & \text{o.w.} \end{cases}$$

This partitions irreducible representations (ρ, V) of G into **equivalence classes**.
i.e. for any irreducible (ρ, V) of G

equivalence class of (ρ, V) = { all (ρ', V') isomorphic to (ρ, V) }

==========

Proof
Let T: V → V' be a non-trivial map
Then for any v ∈ ker T

$T\rho'(g)\, v = \rho(g)\, Tv = \rho(g)\, 0 = 0$

=> $\rho'(g)\, v \in \ker T$
=> ρ' has a subrepresentation on ker T
=> ker T = {0} or V since ρ' is irreducible
=> ker T = {0} since T is non-trivial
i.e. T is injective

Conversely, for any w ∈ Im T

$\rho(g)\, w = \rho(g)\, Tv = T\rho'(g)\, v \in \text{Im } T$ for some w = Tv

=> ρ has a subrepresentation on Im T
=> Im T = V' or {0} since ρ is irreducible
=> Im T = V since T is non-trivial
i.e. T is surjective

Hence T is bijective.

If V = V', then T has an eigvenvalue $\lambda > 0$.
Let W = eigenspace of λ.
Then for any w ∈ W

$T\rho(g)w = \rho(g)\, Tw = \rho(g)\, \lambda w = \lambda\, \rho(g)\, w$

=> $\rho(g)w$ is also an eigenvector
i.e $\rho(g)w \in W$
=> ρ has a subrepresentation in W

=> W = V

QED
==========

Basic properties

Theorem: Basic properties

Let ρ: G → GL(V) be a representation of a finite group G

Eigenvalues Every ρ(g^i) has the **same** eigenvector
i.e. if ρ(g)v = λv, then

ρ(g^i) v = λ^i v

eigenvalue λ^i
eigenvector v

If g has finite order, then λ is a |g|th **root of unity**

λ^|g| = 1 since v = ρ(1)v = ρ(g^|g|)v = λ^|g| v

Hence, the sums **cancel** out each other

$$\sum_k \mathrm{tr}(\rho(g^k)) = \sum_{\lambda} \sum \lambda^k$$

$$= \sum_{\lambda} \sum_k \lambda^k$$

= |g| * (multiplicity of λ = 1)

by sum of roots of unity
i.e. $\sum \lambda^k$ = |g| if λ =1
0 otherwise

Examples

e.g. G = Z4 represented in C* = unit cycle
Let

ρ: G → GL(C*)

Then

ρ(n) ∈ { 1, -1, i, -i } since every n ∈ Z4 has degree 4
i.e. ρ(n)^4 = 1

ρ(1) determines ρ since 1 is a generator of Z4

Hence, possible representations are

$\rho(0) = 1 \quad \rho(1) = i \quad \rho(2) = -1 \quad \rho(3) = -i$

$\rho(0) = 1 \quad \rho(1) = -i \quad \rho(2) = -1 \quad \rho(3) = i$

$\rho(0) = 1 \quad \rho(1) = -1 \quad \rho(2) = 1 \quad \rho(3) = -1$

e.g. Dihedral group Dn

α^n = 1 rotations
β^2 = 1 reflection
$\alpha\beta\alpha\beta = 1$

A representation φ: Dn → GL(R^2) is

$\theta = 2\pi/n$

$\rho(\alpha) = \begin{matrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{matrix}$ e^iθ, e^-iθ

$\rho(\beta) = \begin{matrix} 0 & -1 \\ 1 & 0 \end{matrix}$ 0

Characters are

$\chi\rho(\alpha$^k) = e^ikθ + e^-ikθ = 2cos^2 θ
$\chi\rho(\beta) = 0$

Hence

TODO

==========

====

=== Character theory ==

https://en.wikipedia.org/wiki/Character_theory

Character
A character is simply the **trace tr(ρ(g))** of a representation.
Let ρ: G → GL(V) be a representation of a finite group.

$\chi\rho$: G → F
$\chi\rho$(g) = tr(ρ(g))

If the representation ρ is irreducible, then its character $\chi\rho$ is called **irreducible**.

The first important property of the character is that it it **invariant under similarity**
i.e. since

tr(AB) = tr(BA)

Hence

isomorphism invariant

$\chi\rho$ **=** $\chi\rho$**' if** $\rho \sim \rho$**'**

since if T is an isomorphism,

$\chi\rho'(g)$ = tr(T ∘ ρ(g) ∘ T^-1)
= tr(T^-1 ∘ T ρ(g))
= $\chi\rho$(g)

class function

$\chi\rho$ is a class function
i.e.

$\chi\rho(g^x)$ = $\chi\rho(g)$

since tr(ρ(g^x)) = tr(ρ(x)ρ(g)ρ^-1(x)) = tr(ρ(g))

NOTE: NOT all class functions are representations since class functions are only **semi-multiplicative**

f(gg') = f(gg'^g')
= f(g'g) but not necessarily f(g')f(g)

Due to the additivity of trace, the **decomposition** of representations carries over
i.e.

V = V1⊕ ... ⊕Vn
ρ **=** ρ**1** ⊕ **...** ⊕ ρ**n** where ρ: G → GL(V) ρi: G → GL(Vi)
$\chi\rho$ **=** tr(ρ**1** ⊕ **...** ⊕ ρ**n)** where $\chi\rho$: G → C
= $\chi\rho$**1 + ... +** $\chi\rho$**n**

since tr(A⊕B) = tr(A) + tr(B)

In particular, the **irreducible characters** span all characters.
i.e. let **G^** be the set of all irreducible characters

G^ = { $\chi\xi$**1 ....** $\chi\xi$**n }** where ξi is a representative of each **isomorphism class**

Then

$\chi\rho$ **=** $\sum_{\xi \in G^}$ ρ**^(**ξ**)** $\chi\xi$ where p^(ξ) = multiplicity of $\chi\xi$

i.e.

span { $\chi\xi$**:** $\xi \in$ **G^ }** = all characters $\chi\rho$

<u>Orthonormal basis aka Fourier transform</u>
One of the key property of characters is that the **irreducible characters** form an **orthonormal basis** of not only representations but all **class**

functions.
For any f, h: G → C, define the inner product

$$\mathbf{f \bullet h} = \mathbf{E[f(g)\bar{h}(g)]}$$

$$= 1/|G| \sum_{g \in G} f(g)\, \bar{h}(g)$$

Let L2(G) be the vector space of class functions with the above inner product

f: G → C ∈ L2(G)
f(g^x) = f(g) for any x ∈ G

Then it turns out that the irreducible characters $\chi\xi$ for $\xi \in G\hat{}$ form a **basis** of L2(G)
i.e. each f ∈ L2(G) is a unique linear

$$f = \sum_{\xi \in G\hat{}} \xi^*(f)\, \chi\xi$$ with coefficients aka **dual**
ξ^*: L2(G) → C ∈ L2(G)*

Furthermore, it turns out that the irreducible characters are **orthonormal**.
i.e. for any $\xi, \xi' \in G\hat{}$

$\chi\xi \bullet \chi\xi' = 0$ if $\xi \neq \xi'$ i.e. not isomorphic
$\chi\xi \bullet \chi\xi = 1$

Hence, the duals are precisely the **dot product $\xi^*(f) = f \bullet \chi\xi$**

$$f \bullet \chi\xi = (\sum_{\xi' \in G\hat{}} \xi'^*(f)\, \chi\xi') \bullet \chi\xi$$
$$= \sum_{\xi' \in G\hat{}} \xi'^*(f)\, (\chi\xi' \bullet \chi\xi)$$
$$= \xi^*(f)$$ by orthonormality

aka **Fourier inversion**

This provides a dual of the **magnitude f • f**

$$f \bullet f = E[f(g)\, \bar{f}(g)]$$
$$= E[|f(g)|\hat{}2]$$
$g \in G$

and

$$f \bullet f = f \bullet \sum_{\xi \in G\hat{}} \xi^*(f)\, \chi\xi$$
$$= \sum_{\xi \in G\hat{}} \xi^*(f)\, (f \bullet \chi\xi)$$
$$= \sum_{\xi \in G\hat{}} \xi^*(f)\hat{}2$$

Hence the **dual norm** aka Parseval identity

$$\mathbf{E|f(g)|\hat{}2} = \mathbf{\sum_{\xi \in G\hat{}} \xi^*(f)\hat{}2}$$
$$\mathbf{= f \bullet f}$$

Often written as

$$\mathbf{|| f ||} = \mathbf{|| f^* ||}$$

L2 norm l2 norm

In particular, if f is a character i.e. f = $\chi\rho$, then the Fourier coefficients are exactly the **multiplicity** of the irreducibles
i.e. if

ρ **=** $\oplus$ ρi ρi irreducible

and

kξ = # of ρi ~ ξ

Then

$\chi\rho$ = $\sum_{i}$ $\chi\rho$i
= $\sum_{\xi}$ ($\sum$ 1) $\chi\xi$ over all ρi ~ ξ

= $\sum_{\xi\in G\hat{}}$ kξ $\chi\xi$

Hence

ξ***(**ρ**)** **=** **k**ξ $\in$ **N**

and

$\chi\rho$ $\bullet$ $\chi\rho$ = $\sum$ ξ*(ρ)^2

= $\sum_{\xi\in G\hat{}}$ **k**ξ**^2**

In fact, this is an exact condition for when a class function is a character
Class function f: G → C is a **character** for some representation if and only if the coefficients ξ*(f) are all **integers**.
i.e.

f **is a character** <=> ξ*(f) $\in$ N for all ξ since =>) if ρ = $\oplus$ ni ξi, then $\chi\rho$ = $\sum$ ni $\chi\xi$i
i.e. f = $\chi\rho$ ni = $\chi\rho$ $\bullet$ χi = ξ*(ρ)
<=) let ρ = $\oplus$ ξ*(f) ξ

Then $\chi\rho$ = $\sum$ ξ*(f) $\chi\xi$

And irreducible characters:

ρ **is irreducible** <=> $\chi\rho$ $\bullet$ $\chi\rho$ = 1 since =>) by orthogonality
<=) 1 = $\chi\rho$ $\bullet$ $\chi\rho$ = $\sum$ |ξ*(ρ)|^2 where each ξ*(ρ) is an integer

Hence, ξ0*(ρ) = 1 for exactly one ξ0 and

ρ

$= \sum \xi 0^*(\rho)\ \xi = \xi 0$

Identity 1G: G → C

The identity representation is a special type of **irreducible** representation.

$\sum_{g \in G} \lvert\chi\xi(\mathbf{g})\rvert^2$	$= \lvert G\rvert\ E[\ \lvert\chi\xi(g)\rvert^2\ ]$		
	$= \lvert G\rvert\ \chi\xi \bullet \chi\xi$		
	$= \mathbf{\lvert G\rvert}$		by orthogonality
$\sum_{g \in G} \chi\xi(\mathbf{g})$	$= \lvert G\rvert\ E[\chi\xi(g)]$		
	$= \mathrm{tr}(\sum \xi(g))$		
	$= \lvert G\rvert$	if $\chi = 1$	See basic properties below for proof
	0	o.w.	

i.e. the characters $\chi\xi$ cancel out in the average of $\chi\xi(g)$ but not in the norm $|\chi\xi(g)|$

Hence

representation	$\sum \mathbf{1(g)}$	$= \mathbf{\lvert G\rvert}$
character	$\sum \chi\mathbf{1(g)}$	$= \mathbf{\lvert G\rvert}$

Hence, for any class function $f \in L2(G)$, the average is determined **completely by 1**.

$$\sum_{g \in G} f(\mathbf{g}) = \sum_{g \in G} \sum_{\xi \in G^\wedge} \xi^*(f)\ \chi\xi(g)$$

$$= \sum_{\xi} \xi^*(f) \sum_{g} \chi\xi(g)$$

$= 1^*(f)$ all 0 except for $\xi=1$

$\mathbf{= f \bullet 1}$

Regular representation Reg: G → GL(C[G])

Consider the trace of R(g): G → GL(C[G])
For each basis $eh \in C[G]$ $h \in G$

$$R(g)\ eh = egh$$

Hence R(g)'s matrix has 1 in **(eg, egh) position**, and 0 for all other (egh, _) positions
In particular, the **diagonals** are all 0 except $g \neq 1$
Hence

$\chi R(g)$	$= \mathrm{tr}(R(g))$	
	$= 0$	sum of diagonals = 0
$\chi R(1)$	$= \lvert G\rvert$	

Hence the coefficients are exactly

$\chi R \bullet \chi\xi = 1/|G| \sum \chi R(g)\, \chi\xi(g)$

$= \chi\xi(1)$
$=$ **dim V**$\xi > 0$ where representation $(\xi, V\xi)$

The regular representation thus contains **all irreducible characters**

χ**R** $= \sum \chi\xi(1)\, \xi$
= $\sum$ **dim V**$\xi\ \ \xi$

as expected, since R is the freest representation.

$\sum \chi R(g)$

==========
Proofs

*) Orthogonality

$\chi\rho \bullet \chi\rho' = E[\chi\rho(g)\chi\rho'^{-}(g)]$

$= E[\chi\rho\otimes\rho'^{\wedge}-1\ (g)]$

$= \dim \mathrm{Fix}(\rho\otimes\rho'^{\wedge}-1)$

If $\rho \neq \rho'$, then Fix $(\rho\otimes\rho'^{\wedge}-1) = \varnothing$ since ρ and ρ'

TODO

If $\rho = \rho'$ then

$\rho(g)\otimes\rho(g)^{\wedge}-1\ v\otimes w = \sum$

*) Basis

*) **average** $\sum \chi\xi$
Recall that

$P: G \to GL(V)$
$P = \sum \rho(g)$

is a projection into $\mathrm{Fix}(\rho) = \{ v \in V: \rho(g)v = v \}$.
Hence the eigenvectors are precisely

$Pv = v \quad v \in \mathrm{Fix}(\rho)$

$\Rightarrow \mathrm{tr}(P) = \sum 1$
$= \dim \mathrm{Fix}(\rho)$

But

$\rho(g)v = v \in Fix(\rho)$

=> $\rho|Fix(\rho)$ is a subrepresentation

=> $Fix(\rho) = \varnothing$ if ρ is irreducible, except $\rho=1$

Hence, if $\xi \neq 1$

$\sum \chi\xi$ = tr(P)
= dim Fix(ξ)
= 0

QED
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Theorem: Basic properties

Basic

Identity $\chi\rho(1)$ = dim V

inverse $\chi\rho(g^{-1})$ = $\chi\rho(g)^{-1}$
= $\chi\rho(g)^{-}$ since the eigenvalues are roots of unity
tr($\rho(g)^{-1}$) = $\sum 1/\lambda = \sum \lambda^{-} = tr(\rho(g))^{-}$

sum $\chi\rho + \chi\rho'$ = tr(ρ1)+tr(ρ2)
= tr($\rho 1 \oplus \rho 2$) since tr(A⊕B) = $\sum$ aii + $\sum$ bjj + tr(A) + tr(B)
= $\chi(\rho 1 \oplus \rho 2)$

product $\chi\rho \cdot \chi\rho'$ = tr(ρ) tr(ρ')
= tr($\rho \otimes \rho'$) since tr(A⊗B) = $\sum$i,j aii bjj = tr(A) tr(B)
= $\chi(\rho \otimes \rho')$

Average

E[$\chi\rho$(g)] = dim Fix(ρ)
g∈G

where $Fix(\rho) = \{ v: \rho(g)v = v \text{ for all } G \}$
Note also that

(ρ, Fix(ρ))

is a subrepresentation since $\rho(g)v = v \in Fix(\rho)$ for any $v \in Fix(\rho)$

Hence, if ρ is **irreducible**, then either

E[χ1(g)] = 1
E[$\chi\rho$(g)] = 0 if $\rho \neq 1$ since Fix(ρ) must be $\varnothing$ by irreducibility

invariance

Invariant under **conjugation**
i.e. characters are class functions

$\chi\rho(g) = \chi\rho(g^x)$

Invariant under **isomorphism**
i.e. if $\rho \sim \rho'$, then

$\chi\rho = \chi\rho'$

Irreducible characters
Let ξ be irreducible representations

orthonormal	$\chi\xi \bullet \chi\xi'$	$= 0$ if $\xi \neq \xi'$
	$\chi\xi \bullet \chi\xi$	$= 1$
average	$\sum_{g\in G} \chi\xi(g)$	$= 0$ if $\xi \neq 1$
	$\sum_{g\in G} \chi 1(g)$	$= \lvert G\rvert$
L2 norm	$\sum_{g\in G} \lvert\chi\xi(g)\rvert^2$	$= \lvert G\rvert\ \chi\xi \bullet \chi\xi$
		$= \lvert G\rvert$

Dual sums over G^

dual orthonormal $\sum_{\xi\in G^} \chi\xi(g)\chi\xi^{-}(g') = 0$ if $g \not\sim g'$ by conjugacy classes i.e. $g' \notin g^G$

$\sum_{\xi\in G^} \chi\xi(g)\chi\xi^{-}(g') = \sum \lvert\chi\xi(g)\rvert = \lvert C(g)\rvert$ if $g \sim g'$

note: C(g) = center of g = { x: gx = xg }

note: this is just the function $\varphi(x) = 1$ for a single conjugacy class $x\in g^G$ and 0 o.w.

$$\varphi = \sum \xi^*(\varphi)\ \chi\xi = 1/\lvert C(g)\rvert \sum \xi^{-}(g)\ \chi\xi$$

Orthonormal Basis
Irreducible characters form a basis of class functions

Basis

$f = \sum \xi^*(f)\ \chi\xi = \sum (f \bullet \chi\xi)\ \chi\xi$

where

$\xi^*: G \to C \quad \in G^*$
$\xi^*(f) = f \bullet \chi\xi$

Thus the dimension of class functions L2(G) is

dim (L2(G)) = |G^|
= # of conjugacy classes

Parseval

$E_{g \in G}[|f(g)|^2] = f \bullet f$

$= \sum_{\xi \in G^} (f \bullet \chi\xi)^2$

Characterization
If

$\rho = \oplus c\xi\ \xi$

then

$\chi\rho \bullet \chi\rho = \sum c\xi^2$

where

$\chi\rho \bullet \chi\xi = c\xi \in N$

Hence the characterization

f is a character <=> $f \bullet \chi\xi \in N$ for all ξ

$\rho \in G^$ <=> $\chi\rho \bullet \chi\rho = 1$

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Proofs

*) Orthogonality

$\chi\rho \bullet \chi\rho' = E[\chi\rho(g)\chi\rho'^{-}(g)]$

$= E[\chi\rho\otimes\rho'^{-1}(g)]$

$= \dim \mathrm{Fix}(\rho\otimes\rho'^{-1})$

If $\rho \neq \rho'$, then Fix ($\rho\otimes\rho'^{-1}$) = $\varnothing$ since ρ and ρ'

TODO

If $\rho = \rho'$ then

$\rho(g)\otimes\rho(g)^{-1}\ v\otimes w = \sum$

*) Basis

*) **average** $\sum \chi\rho(g)$
Let

$P: G \to GL(V)$

$P = \sum \rho(g)$

Recall that P is a projection into $Fix(\rho) = \{ v \in V: \rho(g)v = v \}$.
Then

$$\sum \chi\rho(g) = tr(\sum\rho(g)) = tr(P)$$

Hence, it suffices to compute the trace of P.
The eigenvectors are precisely

$$Pv = v \quad v \in Fix(\rho)$$

$$\Rightarrow tr(P) = \sum 1 = \dim Fix(\rho)$$

QED

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== OLD ==

Characters are invariant under **isomorphism**.
i.e. if ρ' is isomorphic to ρ, then

$$\chi\rho'(g) = tr(T \circ \rho(g) \circ T^{-1}) = tr(T^{-1} \circ T\, \rho(g)) = \chi\rho(g) \quad \text{where } \rho' \circ T = T \circ \rho$$

Characters are **class functions** since traces are similarity invariant.
note: a class function is **invariant under conjugation** f(g^x) = f(g) for any x $\in$ G
i.e. for any x $\in$ G

$$\chi\rho(g^x) = tr(\rho(g^x)) = tr(\rho(x)\,\rho(g)\,\rho^{-1}(x)) = \chi\rho(g)$$

by similarity invariance tr(B^-1AB) = tr(A)

Not all class functions are representations since class functions are only **semi-multiplicative**

$$f(gg') = f(gg'^{g'}) = f(g'g)$$

but not necessarily f(g')f(g)

Due to the trace being **additive** tr(A+B) = tr(A) + tr(B) and **semi-multiplicative** tr(AB) = tr(BA)

properties of representations carry over to the character:

arithmetic

additive	$\chi\rho+\chi\rho'$	$= \chi(\rho\oplus\rho')$
semi multiplicative	$\chi\rho\cdot\chi\rho'$	$= \chi(\rho\otimes\rho')$

orthonormal

if (ξ, V), (ξ', V') are irreducible representations of G, then

$$\chi\xi \bullet \chi\xi' = \begin{cases} 1 & \text{if } \xi = \xi' \\ 0 & \text{o.w.} \end{cases}$$

decomposition

if $\rho = \rho 1 \oplus ... \oplus \rho n$ where $(\rho i, Vi)$ are representations then

$$\chi\rho = \chi\rho 1 + ... + \chi\rho n \qquad \text{since } \chi\rho+\chi\rho' = \chi(\rho\oplus\rho')$$

Hence if ξi is a set of all irreducible representations, then every representation has the form

$$\chi\rho = \sum i\ (\chi\rho \bullet \chi\xi i)\ \ \chi\xi i$$

$\chi R(g)$	$= 0$
$\chi R(1)$	$= \lvert G\rvert$
$E[\chi R(g)]$	$= 1$

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proof:

- Average $E[\chi\rho(g)]$

Let P be the average map

P: V → V

P = 1/|G| ∑ ρ(g)

Then

$$E[\chi\rho(g)] = 1/|G| \sum tr(\rho(g)) = 1/|G|\ tr(P)$$

claim: P is a projection into $Fix(\rho)$.

i.e. P^2 = P

If the claim is true, then

$$tr\ P = \dim (Im\ P) = \dim Fix(\rho)$$

Proof claim.
For any $v \in Fix(\rho)$

$$\begin{aligned} Pv &= 1/|G| \sum \rho(g)\, v \\ &= 1/|G| \sum v \\ &= v \end{aligned}$$

$$\Rightarrow Fix(\rho) \subseteq Im\ P$$

Conversely, for any v

$$\begin{aligned} \rho(g')\, Pv &= 1/|G| \sum \rho(g'g)\, v \\ &= Pv \end{aligned}$$

$$\Rightarrow Im\ P \subseteq Fix(\rho)$$

Hence $Im\ P = Fix(\rho)$.
Furthermore, by above

$$Pv = v \quad \text{for any } v \in Fix(\rho)$$
$$\Rightarrow PPv = Pv \quad \text{since } Pv \in Im\ P = Fix(\rho)$$

Hence **P^2 = P** i.e. a projection
QED claim

QED
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